

Josue Torres-Fonseca

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EDUCATION

Doctor of Philosophy in Computer Science and Engineering	Expected May 2028
University of Michigan, Ann Arbor, MI Advisor: Dr. Joyce Chai	GPA: 4.0
Master of Science in Computer Science and Engineering	Graduated May 2025
University of Michigan, Ann Arbor, MI	GPA: 4.0
Bachelor of Science in Computer Science, <i>Minor: Applied Mathematics</i>	Graduated May 2023
Boise State University, Boise, ID	GPA: 3.889, <i>Magna Cum Laude</i> , Dean's List

EXPERIENCE

PhD Candidate, <i>Situated Language and Embodied Dialogue Group</i>	Aug 2023 – Present
University of Michigan, supervised by Dr. Joyce Chai	

- Project 2:
 - Designed and built an embodied conversational robot that encourages individuals living with cognitive decline to reflect on their resilience, co-designed with 10 dementia experts and 14 people with cognitive impairment
 - Developed a structured dialogue manager that steps through rapport-building, challenges, and resilience reflection phases; blinded raters scored its pacing above end-to-end LLM prompting in 35 of 40 simulated conversations
 - Fine-tuned Qwen3-VL-32B on 15K dialogue turns distilled from Gemini 2.5 Pro, plus a resilience topic classifier reaching 84% accuracy; served both on-premise with vLLM to protect participant privacy
 - Deployed in 7 lab and 3 in-home studies with individuals living with cognitive decline; submitted as co-first author to the ACM/IEEE Conference on Human-Robot Interaction (HRI) 2027
- Project 1:
 - Created a safety-centric dataset using PDDL to generate trajectories that complete household tasks while mitigating six categories of kitchen hazards across 30 AI2-THOR scenes
 - Evaluated 11 state-of-the-art VLMs on hazard recognition and safe embodied planning, finding models correctly recognized hazards up to 92% of the time but fixed them less than 60% of the time
 - Developed a decoupled multi-agent framework separating hazard recognition from task execution, raising the rate at which Qwen3-VL-32B fixed appliance-misuse hazards from 0.7% to 71.1%
 - Open-sourced code and dataset; presented a poster at the Association for Computational Linguistics (ACL) 2026

Undergraduate Researcher, <i>Speech Language and Interactive Machines Group</i>	Aug 2021 – Aug 2023
Boise State University, supervised by Dr. Casey Kennington	

- Project 2:
 - Designed and built an incremental spoken dialogue system capable of transcribing and understanding speech, detecting and gaining a concrete understanding of objects, and mapping objects using the Anki Cozmo robot
 - Approximated object permanence by extending Cozmo's SLAM map to remember each object it has seen and its location; in an A/B study with 24 participants, this raised the robot's accuracy answering what/where questions about learned objects from 70.0% to 82.7% and nearly doubled the questions participants asked
 - Presented at Special Interest Group on Discourse and Dialogue (SIGdial) 2022
- Project 1:
 - Collected and analyzed HADREB, a dataset of over 1,000 robot emotional behaviors described and rated by 32 participants on the Misty II and Cozmo robots
 - Showed a zero-shot language model predicts perceived emotion from the descriptions with up to 79.3% accuracy
 - Presented at Language Resources and Evaluation Conference (LREC) 2022

Undergraduate Researcher, <i>Language & Intelligence Group</i>	May 2022 – Nov 2022
Massachusetts Institute of Technology, supervised by Dr. Jacob Andreas	

- Designed and built an interpretable ML model providing neuron annotations using no data
- Presented a poster and delivered a lightning talk at the end of the MIT Summer Research Program

Software Engineer Intern

May 2021 – Aug 2021

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- Designed, developed, tested, and deployed new features in project releases as part of a 13-member team
- Applied Agile and Scrum; gained full-stack experience with JavaScript, React/Redux, C#, .NET, SQL, and Git

Undergraduate Researcher, *Intelligent Security Group*

Jun 2020 – Aug 2020

Texas State University, supervised by Dr. Mina Guirguis (NSF Smart & Connected Communities REU)

- Designed and built a game and reward system to secure a simulated water tank using game theory

PUBLICATIONS

Robot to Elicit Resilience Reflection for Individuals Living with Cognitive Decline

Under Review

Long-Jing Hsu*, Josue Torres-Fonseca*, Janice Bays, Joyce Chai, and Patricia Alves-Oliveira. 2027. *Robot to Elicit Resilience Reflection for Individuals Living with Cognitive Decline*. Under review at the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027). (*Equal contribution, co-first authors)

SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models

Jul 2026

Josue Torres-Fonseca, Naihao Deng, Yinpei Dai, Shane Storks, Yichi Zhang, Rada Mihalcea, Casey Kennington, Joyce Chai. 2026. *SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models*. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 37190-37211, San Diego, CA, USA. Association for Computational Linguistics.

The Mechanistic Emergence of Symbol Grounding in Language Models

Jul 2026

Shuyu Wu, Ziqiao Ma, Xiaoxi Luo, Yidong Huang, Josue Torres-Fonseca, Freda Shi, Joyce Chai. 2026. *The Mechanistic Emergence of Symbol Grounding in Language Models*. In *Proceedings of the International Conference on Machine Learning 2026*, Seoul, South Korea.

rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms

Aug 2025

Anna Manaseryan, Porter Rigby, Brooke Matthews, Catherine Henry, Josue Torres-Fonseca, Ryan Whetten, Enoch Levandovsky, and Casey Kennington. 2025. *rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms*. In *Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 637–640, Avignon, France. Association for Computational Linguistics.

Symbol and Communicative Grounding through Object Permanence with a Mobile Robot

Sep 2022

Josue Torres-Fonseca, Catherine Henry, and Casey Kennington. 2022. *Symbol and Communicative Grounding through Object Permanence with a Mobile Robot*. In *Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 124–134, Edinburgh, UK. Association for Computational Linguistics.

HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors

Jun 2022

Josue Torres-Fonseca and Casey Kennington. 2022. *HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors*. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 5739–5748, Marseille, France. European Language Resources Association.

HONORS & AWARDS

Gemini Academic Program Award	(\$10,000)	Spring 2026 – Spring 2027
NSF Graduate Research Fellowship	(\$159,000)	Fall 2023 – Spring 2028
Boise State University Top Ten Scholar		Spring 2023
Barry M. Goldwater Scholarship	(\$7,500)	Fall 2022 – Spring 2023
Federal Employee Education & Assistance Fund Scholarship	(\$1,000)	Fall 2018

TEACHING EXPERIENCE

Tutor , <i>Operating Systems</i> , Boise State University	Jan 2023 – May 2023
Tutor , <i>Introduction to Theory of Computation</i> , Boise State University	Jan 2021 – May 2021
Tutor , <i>Data Structures</i> , Boise State University	Aug 2020 – Dec 2020
Lab Assistant , <i>Microprocessors</i> , Boise State University	Aug 2020 – Dec 2020

ORGANIZATIONS

Treasurer , <i>Computer Science and Engineering Graduate Students</i>	Fall 2024 – Spring 2025
President , <i>Artificial Intelligence Club</i>	Fall 2024 – Spring 2025
President , <i>Competitive Cybersecurity and Cyber-Physical Club</i>	Fall 2019 – Spring 2022